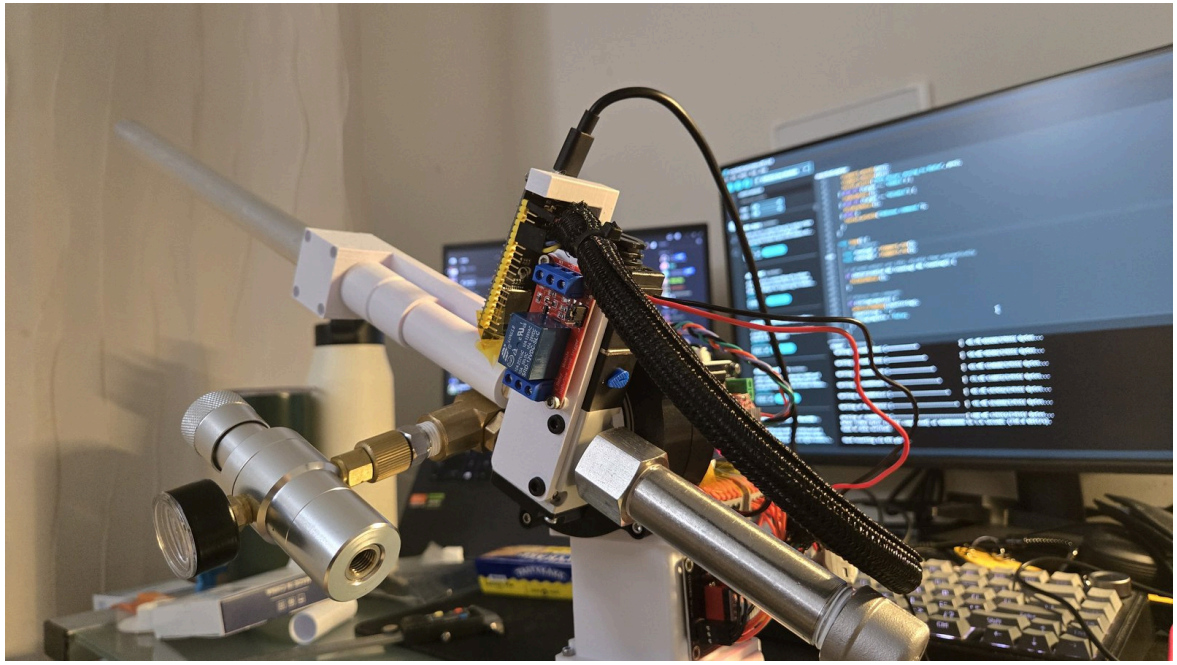


Autonomous Tracking Turret

A two axis turret that finds a person on camera, keeps them centered, and fires foam darts with CO2.

I designed and built the hardware for a turret that tracks a person and fires foam darts. Direct drive steppers could not move the top heavy head, so I designed printed cycloidal reducers for both axes. An ESP32 streams video to a laptop that runs pose detection and sends aim corrections back over UDP.

Context	Dates	My role
Personal project	Summer 2025	Mechanical design, electronics, firmware
Partner	**Tools**	**Outcome**
Jinson Jimmy, University of Maryland (pose tracking software)	SolidWorks, Desmos, ESP32 (Arduino C++), Python	Working tracking and firing prototype, under $150

12:1 and 18:1

printed cycloidal reducers on NEMA 17 steppers

97%

of peak modeled dart speed from a 0.3 m barrel

2 weeks

hardware build for under $150

Project	Sheet title	Engineer	Date	Sheet
Autonomous Tracking Turret	Overview	Xiang Wang	Summer 2025	1 of 6

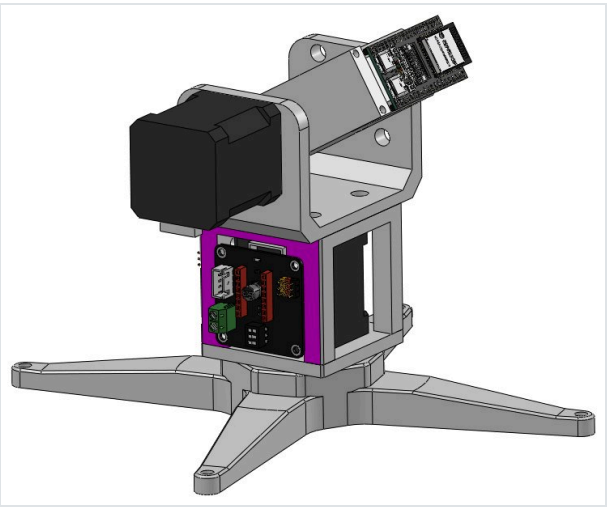
Requirements and first prototype

The goal was a turret that points a launcher at a moving person automatically, on a student budget. From the first brainstorm I fixed four choices: two axes, CO2 instead of springs, stepper motors, and an ESP32 instead of a Raspberry Pi. Each one was picked for cost and simplicity. The base motor rotates with the head so the turret can sweep a full 360 degrees without twisting its wiring.

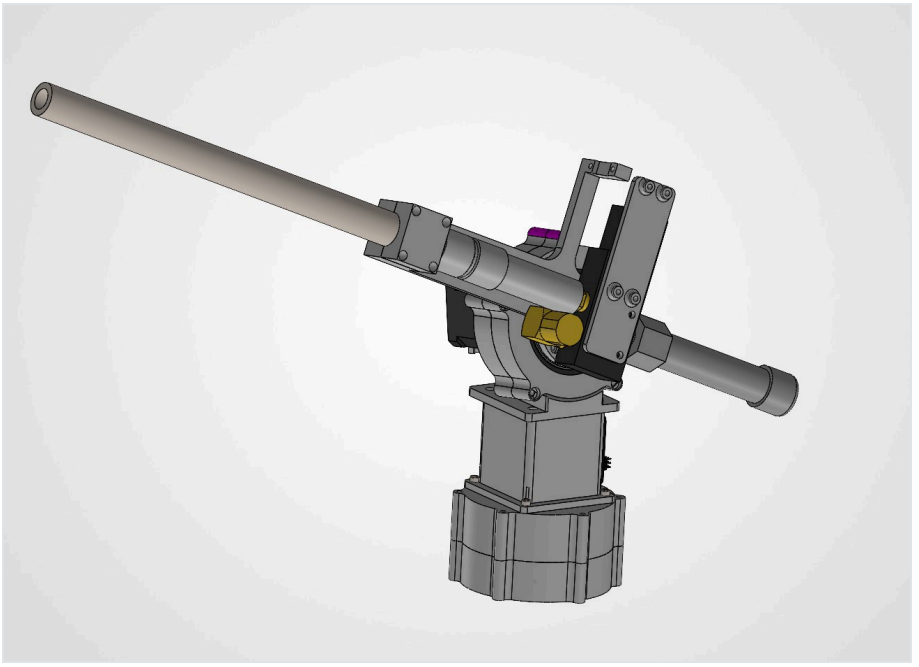
The first prototype drove both axes directly. It spun freely with no load, but the base NEMA 17 could not turn the finished head. The fix had to add torque without adding much size or a larger motor.



2.1 First prototype: direct drive, before reduction



2.2 First prototype CAD: both axes direct drive



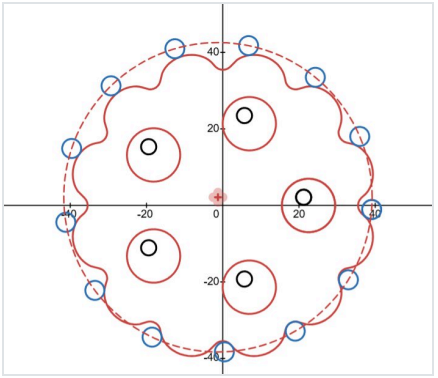
2.3 Final CAD: reducers on both axes, barrel, and gas system

Project	Sheet title	Engineer	Date	Sheet
Autonomous Tracking Turret	Requirements and first prototype	Xiang Wang	Summer 2025	2 of 6

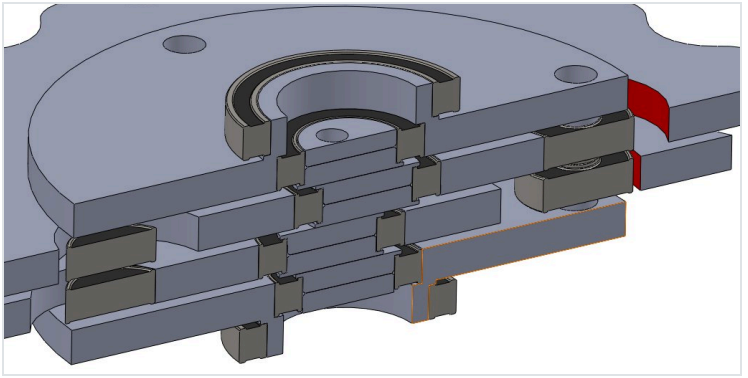
Cycloidal reducers

A cycloidal drive gives a high ratio in a single compact stage and runs quietly, and every part can be printed. I derived the cycloid disc profile parametrically in Desmos, adjusting pin count, pin radius, and eccentricity, then brought the curve into SolidWorks as the disc sketch.

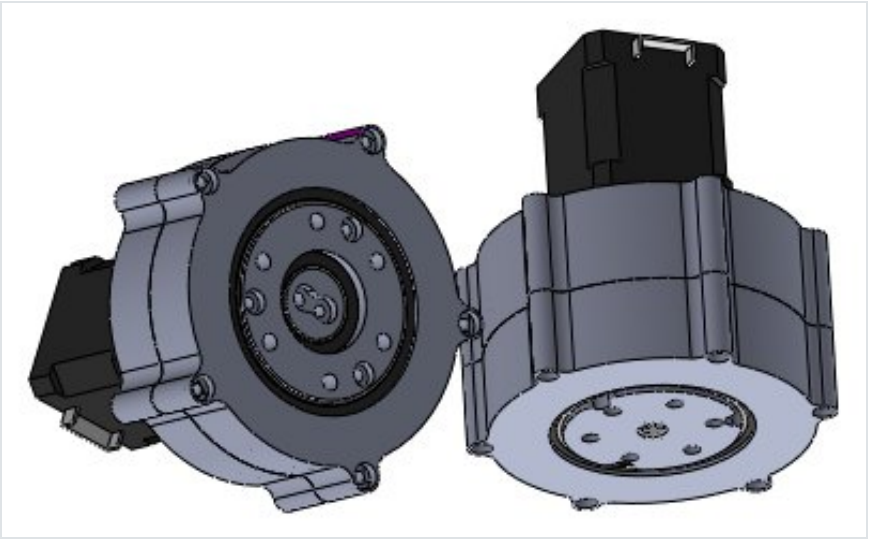
The result was two drives: 18:1 for azimuth and 12:1 for elevation, both on NEMA 17 steppers. I checked each printed drive by hand for smooth engagement before mounting it, then designed the brackets that tie the two axes together.



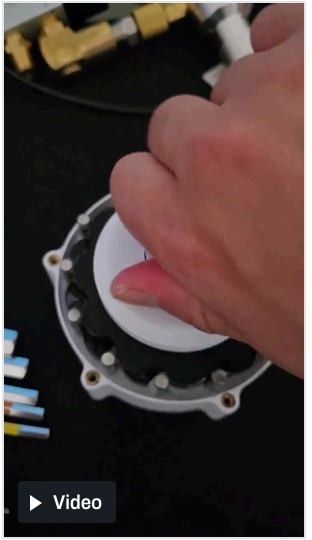
3.1 Disc profile and pin circle generated in Desmos



3.2 Section through the reducer stack



3.3 Elevation and azimuth drives



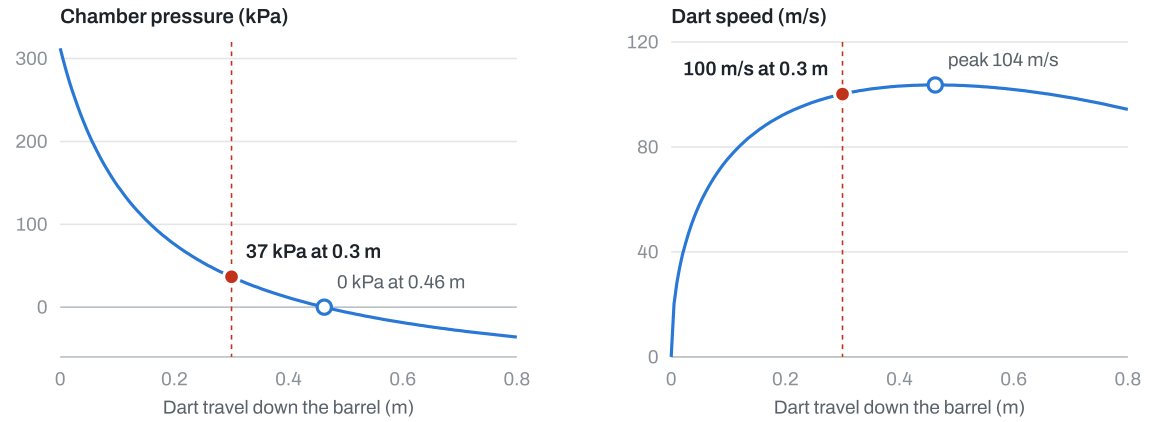
3.4 Printed drive turning by hand

Project	Sheet title	Engineer	Date	Sheet
Autonomous Tracking Turret	Cycloidal reducers	Xiang Wang	Summer 2025	3 of 6

CO2 firing system

Gas path: CO2 cartridge, regulator, a small shot chamber, a 3/2 way solenoid valve, then the barrel. On the bench the valve clicked at 12 V but never shifted, and raising the voltage changed nothing. Cycling it by hand only opened one set of ports. The cause was supply pressure: the valve will not shift below about 20 psi, which is typical of pilot operated valves, so it could never cycle on an unpressurized bench.

To size the chamber and barrel I modeled the shot as isothermal expansion pushing a 1 g dart. At 60 psi the model predicts about 100 m/s after 0.3 m of travel, which is 97% of the model's peak speed, with 37 kPa still in the chamber. A longer barrel adds almost nothing. The model ignores friction, so the real setting needs some margin. An early live shot broke the dart outright.



Shot chamber model: pressure and dart speed against travel, 60 psi, 1 g dart. Red marks the 0.3 m barrel.



4.1 Gas path laid out before mounting



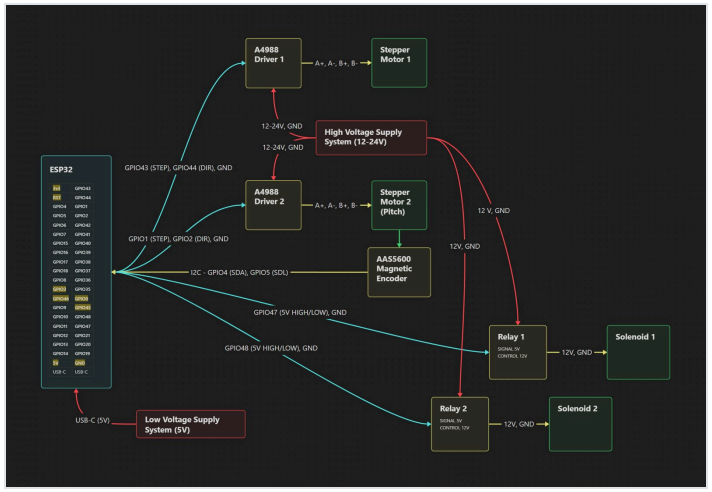
4.2 An early test shot broke the dart

Project	Sheet title	Engineer	Date	Sheet
Autonomous Tracking Turret	CO2 firing system	Xiang Wang	Summer 2025	4 of 6

Electronics and firmware

An ESP32 with a camera runs everything on the turret. Two A4988 drivers on a 12 to 24 V rail drive the steppers, an AS5600 magnetic encoder reads the pitch axis over I2C, and relays switch the 12 V solenoid valves. Before wiring I audited every ESP32 pin for strapping, UART, and USB conflicts so nothing would fight the boot sequence.

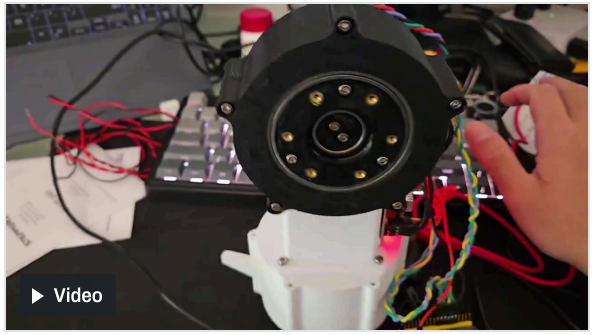
Tracking is split across two machines. The ESP32 streams video over WiFi and executes motion and firing. A laptop runs MediaPipe pose detection, written by Jinson, and sends aim corrections over UDP. The firmware turns each correction into stepper moves with AccelStepper.



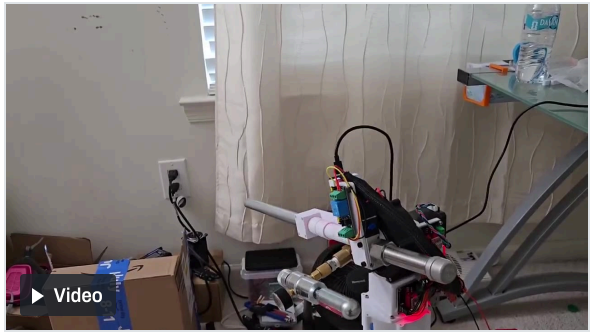
5.1 System wiring

Pin name	Pin number	HIGH Signal?	Voltage	Safe for GPIO	Notes
G14	14	☑	3.3	SAFE	
G13	13	☑	3.3	SAFE	
G12	12	☑	3.3	SAFE	
G11	11	☑	3.3	SAFE	
G10	10	☑	3.3	SAFE	
G9	9	☑	3.3	SAFE	
G46	46	☐	0	UNSAFE	Bootstrapping (strapping pin)
G3	3	☐	0	UNSAFE	UART0 RX (programming)
G8	8	☑	3.3	SAFE	
G18	18	☑	3.3	SAFE	
G17	17	☑	3.3	SAFE	
G16	16	☑	3.3	SAFE	
G15	15	☑	3.3	SAFE	
G7	7	☑	3.3	SAFE	
G6	6	☑	3.3	SAFE	
G5	5	☑	3.3	SAFE	
G4	4	☑	3.3	SAFE	
G1	1	☐	0	UNSAFE	UART0 TX (programming)
G2	2	☐	0	SAFE	
G42	42	☑	3.3	SAFE	
G41	41	☑	3.3	SAFE	
G40	40	☑	3.3	SAFE	
G39	39	☑	3.3	SAFE	
G38	38	☑	3.3	SAFE	
G37	37	☑	3.3	SAFE	
G36	36	☑	3.3	SAFE	
G35	35	☑	3.3	SAFE	
G0	0	☐	4.3	UNSAFE	Bootstrapping (strapping pin)
G45	45	☑	3.3	SAFE	
G48	48	☑	3.3	SAFE	
G47	47	☑	3.3	SAFE	
G21	21	☑	3.3	SAFE	
G20	20	☐	3.3	UNSAFE	USB D+
G19	19	☐	0	UNSAFE	USB D-

5.2 ESP32 pin audit: green pins are safe to drive at boot



5.3 Both axes moving under firmware control



5.4 Motion and fire sequence (gas off to save cartridges)

Project	Sheet title	Engineer	Date	Sheet
Autonomous Tracking Turret	Electronics and firmware	Xiang Wang	Summer 2025	5 of 6

Results and next steps

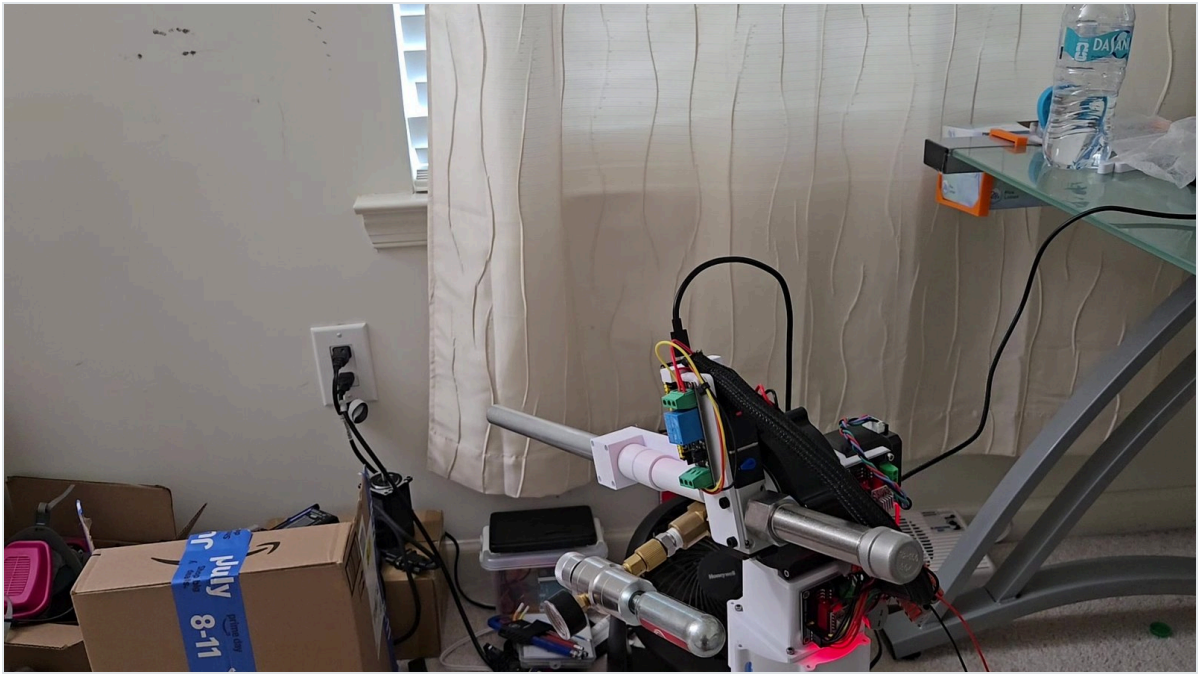
The prototype moved both axes under firmware control, ran the fire sequence, and fired darts on the bench. The limits were in the camera path: the ESP32 was slow to capture and send frames, could not run detection locally, and its camera resolution capped tracking range.

Next steps: a battery so the turret runs untethered, a single board computer to run detection on the turret, and automatic dart feeding.

Files

Case study (PDF)
Design journal (PDF)
18:1 cycloidal drive (STEP)
12:1 cycloidal drive (STEP)
ESP32 firmware
Pose tracking script
github.com/xiang-wang-US/Cycloidal_Turret

Pose tracking software by Jinson Jimmy (University of Maryland).



6.1 Running the motion and fire sequence